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AI Source Localization and Adaptive Logic

APS

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CHECKPOINT

ADAPTIVE PATHOGEN SURVEILLANCE | INDIVIDUAL PROGRESS BRIEF

MY ROLE

I focus on predicting which facility zone is the most likely source of a simulated outbreak. I will compare simple rules with machine-learning models and explore whether an extra sensor reading improves the prediction.

OUR WORKING FOUNDATION

Our shared prototype already has a strongest-sensor baseline and basic source-location error measurements. I am using those as the reference for later model comparisons. The learned classifiers and adaptive sampling are still part of our next development phase unless my own tests confirm otherwise.

Model confidence is an estimated probability, not proof that a prediction is correct. I will interpret it carefully.

MY CHECKPOINT STATUS

I am: [building / testing / refining]

MY CONTRIBUTION SO FAR

I have completed: [specific file, test, or output]

TERMS AND IDEAS I USE

A baseline is a simple method to compare against; ours predicts the source at the strongest available sensor. Logistic Regression learns weighted patterns in sensor readings to classify a source zone. Random Forest combines many decision trees to make a prediction. Exact-source accuracy counts fully correct guesses. Graph-distance error counts how many zone connections separate a predicted source from the real one. Adaptive sampling means requesting one more simulated reading where it may be most useful.

WHAT I AM WORKING ON NEXT

1. I will train Logistic Regression and Random Forest using a consistent dataset and a held-out set of test runs.
2. I will compare the models against the baseline using exact-source accuracy and graph-distance error, including noisy and missing readings.
3. If the classifiers work reliably, I will test one additional sample selected from an uncertain prediction and measure whether it helps.

CHALLENGES I AM CONSIDERING

A model could memorize patterns from our simulator without generalizing to unseen runs. I need to avoid data leakage and check whether extra sampling actually helps rather than assuming it will.

WHAT I PLAN TO SHOW

I plan to show a runnable prediction example and a small comparison of baseline and model results when those tests are available.

HOW I WOULD EXPLAIN MY WORK

My checkpoint summary: My area is the source-location decision. We already have a basic rule to compare against, and I want to test whether trained classifiers can do better with imperfect readings. After that, I will check whether one intelligently chosen extra measurement changes the answer in a useful way.